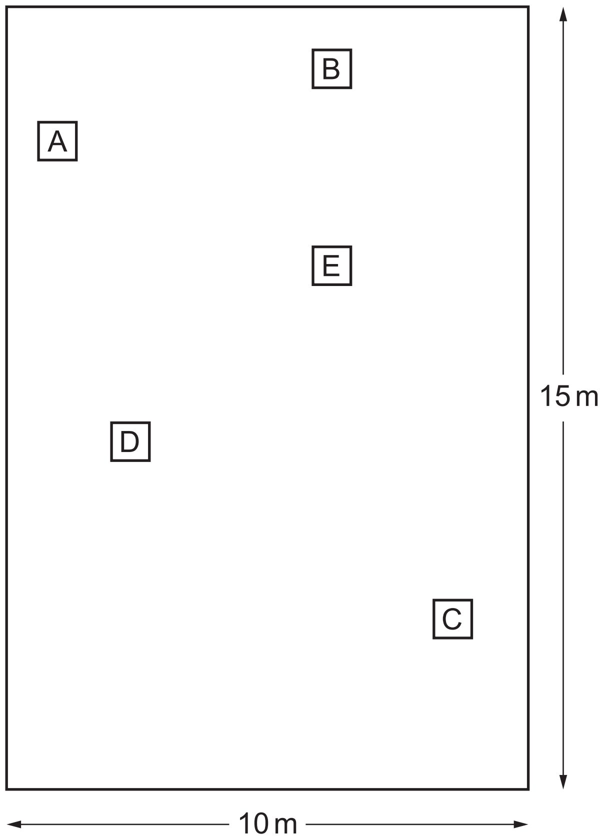
Examiner only

6. Some students investigated the number of dandelion plants on a lawn. The diagram shows the lawn and the location of 5 quadrats **(A to E)** which the students had placed at random.



The students counted the number of dandelions in each quadrat and recorded their results in the table below.

Quadrat	Number of dandelions
A	6
B	3
C	4
D	8
E	4

(a) Calculate the mean number of dandelions per quadrat. [1]

mean number of dandelions = ...

(b) Calculate the area of the lawn. [1]

area of lawn = ... m^2

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(c) Each quadrat has an area of 1 m^2. Calculate how many quadrats will fit in the area of the lawn. [1]

number of quadrats = ...

(d) Use your answers to (a) and (c) to calculate the total number of dandelions on the lawn. [1]

number of dandelions = ...

(e) The actual number of dandelions on the lawn is 600.
Use the formula below, and your answer to (d), to calculate the percentage error of your answer above. [2]

$$\text{percentage error} = \frac{\text{number of dandelions} - 600}{600} \times 100$$

percentage error = ... %

(f) State **one** improvement to the method used by the students which would give more confidence in their answer for the number of dandelions. [1]

....

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